

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 14: Constructions and Tilings

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	14 — Constructions and Tilings
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. A polygon does not tile merely because 360 is 'close' to a multiple of its angle, a triangle is constructible only when the triangle inequality and a valid angle sum hold, and you cannot trisect a general angle the way you bisect it. Do not write on this paper — mark answers on your answer sheet.

1. You must construct the perpendicular bisector of a segment PQ that is 8 cm long, but your compass can open to a maximum radius of only 3 cm. Using arcs of equal radius from P and from Q, can the perpendicular bisector be constructed?

- (A) No, because the arcs need a radius greater than half of PQ (more than 4 cm) so that the two arcs cross on each side; 3 cm is too small and the arcs never meet.
- (B) Yes, any compass radius works as long as it is the same from both ends.
- (C) Yes, because 3 cm is less than the full 8 cm length, which is the only requirement.
- (D) No, because the radius must be exactly half of PQ, namely 4 cm.

2. A perpendicular to a line m is to be erected at a point A lying on m. From A you mark two points equidistant from A on m, call them L and R, then swing equal arcs from L and R that cross above the line at T. The ray AT is perpendicular to m because T is:

- (A) equidistant from L and R, so AT is the perpendicular bisector of segment LR, and that bisector passes through A and is perpendicular to m.
- (B) the midpoint of segment LR, which automatically makes any line through it perpendicular.
- (C) 60° away from the line, which is then doubled to a right angle.
- (D) the centre of a circle through L, R and A, so AT is a radius.

3. Every point on the perpendicular bisector of segment AB is equidistant from A and B. Using only this fact, where must the centre of a circle that passes through BOTH A and B lie?

- (A) Somewhere on the perpendicular bisector of AB, since the centre must be equally far from A and from B.
- (B) Exactly at the midpoint of AB and nowhere else.
- (C) Anywhere inside segment AB.
- (D) On the line AB extended, beyond A or beyond B.

4. The perpendicular bisector of XY meets XY at exactly one angle. A student insists it could be 80° if the compass radius is chosen carefully. What is the truth?

- (A) It is 80° only if the radius is less than the length of XY.
- (B) It can be any angle depending on the compass setting.
- (C) It is 60° , the natural angle of compass arcs.
- (D) It is always exactly 90° ; the equal-radius arcs force a right angle no matter what (valid) radius is chosen.

5. You are asked to construct a triangle with sides 4 cm, 5 cm and 10 cm using SSS. Before drawing, you check feasibility. The triangle:

- (A) can be constructed, since all three lengths are different.
- (B) can be constructed, since 10 is less than $4 + 5 + \text{anything}$.
- (C) cannot be constructed, because $4 + 5 = 9$ is less than 10, violating the triangle inequality (two sides must sum to more than the third).
- (D) cannot be constructed, because no side may exceed 5 cm.

6. To construct a triangle by SAS you are given two sides and the angle BETWEEN them. Which of these data sets is a valid SAS set that determines a unique triangle?

- (A) Sides 6 cm and 5 cm with the 50° angle lying between those two sides.
- (B) Sides 6 cm and 5 cm with a 50° angle that is opposite the 6 cm side.
- (C) Three angles 50° , 60° and 70° , with no side given.
- (D) Sides 6 cm, 5 cm and 12 cm with no angle given.

7. A triangle is to be built by ASA with the two given angles measuring 110° and 80° , on a base of 6 cm. Is this constructible?

- (A) No, because $110^\circ + 80^\circ = 190^\circ$ already exceeds 180° , leaving no room for a third angle, so no triangle exists.
- (B) Yes, the third angle is $180 - 190 = -10^\circ$, which is allowed.
- (C) Yes, because the base length 6 cm is positive.
- (D) No, because ASA requires the two angles to be equal.

8. Using RHS you construct a right triangle given the hypotenuse 13 cm and one leg 5 cm. The construction succeeds and the other leg comes out as:

- (A) 18 cm, since $13 + 5 = 18$.
- (B) 12 cm, since the right angle and Pythagoras give $\text{leg} = \sqrt{(13^2 - 5^2)} = \sqrt{144} = 12$.
- (C) 8 cm, since $13 - 5 = 8$.
- (D) It cannot be built, since the hypotenuse must equal the leg.

9. Which of the following data sets does NOT determine a unique triangle (i.e. more than one non-congruent triangle can fit)?

- (A) Three sides satisfying the triangle inequality (SSS).
- (B) Two sides and an angle that is NOT between them (the SSA / 'ambiguous' case).
- (C) Two angles and the side between them (ASA).
- (D) Two sides and the angle between them (SAS).

10. A regular polygon has interior angle 156° . Using the relation that the interior angle of a regular n-gon is $180^\circ - (360^\circ/n)$, how many sides does it have?

- (A) 12 sides, because $156 \div 13 \approx 12$.
- (B) 15 sides, because $360^\circ/n = 180^\circ - 156^\circ = 24^\circ$, so $n = 360 \div 24 = 15$.
- (C) 10 sides, because 156° is between a decagon and a dodecagon.
- (D) 24 sides, because the exterior angle equals the interior angle minus 132° .

11. Bisecting a given 60° angle is easy with compass and straightedge, but a classmate claims you can TRISECT any angle the same way, by 'just swinging three equal arcs'. The correct response is:

- (A) Trisection is just as easy as bisection; three equal arcs always split the angle into thirds.
- (B) Trisection is possible only for angles larger than 90° .
- (C) Trisecting a general angle is impossible with only compass and straightedge; bisection works, but three equal arcs along an arm do not divide the ANGLE into three equal parts.
- (D) Neither bisection nor trisection is possible without a protractor.

12. To copy an angle ABC onto a new ray, you draw an arc across both arms of ABC cutting them at P and Q, then transfer that same arc onto the new ray and STEP OFF the chord PQ. The construction reproduces the angle exactly because:

- (A) the protractor reading is carried over invisibly by the compass.
- (B) equal radii give congruent arcs, and copying the chord PQ forces the same arc length, hence the same angle, at the new vertex.
- (C) any two arcs of the same colour subtend the same angle.
- (D) stepping off PQ doubles the angle, which is then halved.

13. You are asked to construct a triangle with angles 90° , 90° and 0° . Is this possible?

(A) No, because the three angles of a triangle must sum to 180° and each must be positive; $90 + 90 + 0 = 180$ but a 0° angle means two sides coincide, so it is not a genuine triangle.

(B) Yes, because the angles add to exactly 180° . (C) Yes, it forms a thin right triangle.

(D) No, because the sum 180° is too small for any triangle.

14. A perpendicular is dropped from an external point F to a line m. You swing an arc from F crossing m at two points G and H, then construct the perpendicular bisector of GH. The foot of the perpendicular from F is:

(A) the point G, the first crossing of the arc.

(B) any point on m, since all perpendiculars look the same.

(C) the point on m closest to G, not to F.

(D) the midpoint of GH, where the perpendicular bisector of GH meets m; the segment from F to that midpoint is perpendicular to m.

15. The interior angles of a regular polygon are each 162° . How many of these polygons would meet at a single point if they tiled, and does the polygon tile by itself?

(A) $360 \div 162 \approx 2.22$, round to 2, so two tile at a vertex.

(B) $360 \div 162 \approx 2.22$, which is not a whole number, so it does NOT tile by itself.

(C) It tiles, with about 2.22 polygons at a vertex.

(D) It tiles, because 162° is close to 180° .

16. For a regular polygon to tile the plane by itself, its interior angle A must satisfy which exact condition?

(A) 360° must be an exact whole-number multiple of A (i.e. $360 \div A$ is a positive integer).

(B) A must be less than 120° .

(C) A must be a multiple of 45° .

(D) $360 \div A$ must be at least 2, whether or not it is a whole number.

17. A regular dodecagon has interior angle 150° . At a tiling vertex, ONE dodecagon corner is placed; the remaining angle is filled by ONE more regular polygon. That polygon's interior angle must be:

(A) 150° , another dodecagon.

(B) 60° , an equilateral triangle.

(C) 90° , a square.

(D) 210° , which is impossible for a regular polygon (interior angles are always under 180°), so a single extra polygon cannot complete the vertex.

18. At a vertex of a semi-regular tiling, the corner angles must total exactly 360° . A proposed vertex uses one square (90°), one regular hexagon (120°) and one regular dodecagon (150°). Does this vertex close up exactly?

(A) Yes, because $90 + 120 + 150 = 360$ exactly, so the three polygons fit around the point with no gap or overlap.

(B) No, it leaves a 30° gap.

(C) No, it overlaps by 30° .

(D) Yes, but only because all three angles are multiples of 30° , regardless of the sum.

19. A floor is tiled with regular octagons (135°) and one other regular polygon. At each vertex two octagons meet plus one more tile. The third tile's interior angle, and its name, are:

- (A) 60° — an equilateral triangle.
- (B) 120° — a regular hexagon.
- (C) 90° — a square, since $360 - 2 \times 135 = 360 - 270 = 90$.
- (D) 45° — a regular octagon corner cut in half.

20. Three regular pentagons (108° each) are pushed snugly around a point. After studying the leftover, a student wants to fill it with a single regular polygon. What interior angle would that polygon need, and is it possible?

- (A) It would need 36° (since $360 - 3 \times 108 = 36$), but no regular polygon has a 36° interior angle, so the gap cannot be filled this way.
- (B) It would need 36° , which an equilateral triangle supplies.
- (C) It would need 72° , supplied by a regular pentagon half.
- (D) The leftover is 0° , so three pentagons already tile.

21. While constructing a perpendicular bisector, your compass slipped so that the arc from X used radius 5 cm but the arc from Y used radius 6 cm. The line you draw through the two crossings will:

- (A) still be the exact perpendicular bisector, since both crossings exist.
- (B) be parallel to XY.
- (C) become the angle bisector of angle X instead.
- (D) not be the true perpendicular bisector — it will be tilted and will cross XY off the midpoint, nearer the end you drew with the larger radius having its crossing pushed across.

22. You are given a base $BC = 7$ cm, and told the angle $B = 40^\circ$ and the angle $C = 30^\circ$. The triangle is constructed by ASA. The third angle A and the constructibility are:

- (A) $A = 70^\circ$, and it is constructible.
- (B) $A = 110^\circ$, and the triangle is constructible, since $40 + 30 = 70 < 180$ leaves a valid positive third angle.
- (C) $A = 110^\circ$, but it is not constructible because two angles are unequal.
- (D) It is not constructible, since the base must be the longest side.

23. Among regular polygons, the self-tiling ability stops after the hexagon. The precise reason is that, beyond six sides:

- (A) the interior angle exceeds 180° , which cannot occur in a tiling.
- (B) the polygons become too large to fit on a floor.
- (C) the exterior angle becomes negative.
- (D) the interior angle exceeds 120° but stays below 180° , so 360° divided by it falls strictly between 2 and 3 and is never a whole number.

24. A 60° angle is constructed, then bisected to 30° , then that 30° is bisected, and that result is bisected once more. The smallest angle now present measures:

- (A) 15° , after two halvings.
- (B) 7.5° , since $60 \rightarrow 30 \rightarrow 15 \rightarrow 7.5$ after three successive halvings.
- (C) 10° , since $60 \div 6 = 10$.
- (D) 3.75° , after four halvings.

25. A regular polygon has interior angle 165° . A student computes $360 \div 165 \approx 2.18$ and concludes 'round it to 2, so two tile at a vertex'. The flaw is:

- (A) the division is correct and two really do tile with a tiny harmless gap.
- (B) two such corners give $2 \times 165 = 330^\circ$, which is 30° short of 360° , leaving a gap; rounding the divisor does not produce a real tiling.
- (C) they should have rounded up to 3, giving a perfect tiling.
- (D) the polygon does tile, because 165° is under 180° .

26. A regular hexagon's interior angle is 120° . In a mixed tiling, at one vertex you place one hexagon (120°) and want to fill the rest with equilateral triangles (60° each). How many triangles complete the vertex exactly?

- (A) Two triangles, since $360 - 120 = 240$ and $240 \div 120 = 2$.
- (B) Three triangles, since $3 \times 60 = 180$ fills the rest.
- (C) Four triangles, since $360 - 120 = 240$ and $240 \div 60 = 4$.
- (D) Six triangles, since six triangles always meet at a point.

27. To construct a 105° angle with compass and straightedge, which combination of constructible angles works?

- (A) $60^\circ + 30^\circ + 20^\circ$, since $60 + 30 + 20 = 110$.
- (B) $90^\circ + 30^\circ$, since that equals 120° .
- (C) trisecting a 315° angle, giving 105° directly.
- (D) $60^\circ + 45^\circ$, where 45° is half of a constructed 90° , since $60 + 45 = 105$.

28. Two squares (90° each) and three equilateral triangles (60° each) are arranged around a point. Do they close the vertex exactly?

- (A) No, it leaves a 30° gap.
- (B) No, it overlaps by 30° .
- (C) Yes, but only because there are five tiles in total.
- (D) Yes, because $2 \times 90 + 3 \times 60 = 180 + 180 = 360$ exactly.

29. A rhombus tile is made and you ask whether ANY triangle (not just equilateral) can tile the plane. The correct general fact is:

- (A) Every triangle tiles the plane, because two copies of any triangle join to form a parallelogram, and parallelograms tile.
- (B) Only the equilateral triangle tiles; other triangles leave gaps.
- (C) No triangle tiles unless its angles divide 360° evenly.
- (D) Only right triangles tile, by forming rectangles.

30. At a tiling vertex you find squares and triangles arranged so the angles total 350° . This means:

- (A) the vertex does not close — there is a 10° gap, so it is not a valid tiling vertex.
- (B) the tiling is valid, since 350° is close enough to 360° .
- (C) there is a 10° overlap.
- (D) the angles must be remeasured, because under- 360° is impossible.

31. You bisect a given angle ABC by drawing an arc from B cutting the arms at P and Q, then equal arcs from P and Q meeting at R. The ray BR bisects the angle because R is:

- (A) the midpoint of segment PQ only, which need not bisect the angle.
- (B) exactly 30° from each arm regardless of the original angle.
- (C) the centre of the circle through P, Q and B.
- (D) equidistant from the two arms (equal distances from P and Q on equal arms), so BR splits angle ABC into two equal halves.

32. A regular polygon tiles the plane with k copies meeting at each vertex. If $k = 3$, the interior angle of the polygon must be:

- (A) 60° , the equilateral triangle.
- (B) 90° , the square.
- (C) 120° , since $3 \times (\text{interior angle}) = 360$, giving 120° — the regular hexagon.
- (D) 108° , the regular pentagon.

33. A semi-regular tiling vertex is built from a regular 12-gon (150°) and two equilateral triangles (60° each) plus one more polygon. The fourth polygon's interior angle is:

- (A) 60° , another equilateral triangle.
- (B) 90° , since $360 - 150 - 60 - 60 = 90$ — a square.
- (C) 120° , a regular hexagon.
- (D) 150° , another 12-gon.

34. You construct the perpendicular bisectors of two different sides of a triangle. The point where they cross is special because it is:

- (A) equidistant from all three sides — the centre of the inscribed circle.
- (B) equidistant from all three vertices — the centre of the circle passing through all three vertices (the circumcentre).
- (C) always the midpoint of the longest side.
- (D) the point where the three angle bisectors meet.

35. A regular polygon has exterior angle 40° . Its interior angle, and whether it tiles by itself, are:

- (A) interior 140° ; it tiles with 2 corners (rounding 2.57 down).
- (B) interior 320° ; it tiles because 320 is large.
- (C) interior 140° ; it does not tile, since $360 \div 140 \approx 2.57$ is not a whole number.
- (D) interior 140° ; it tiles with 3 corners.

36. Tiling (tessellation) of the plane is defined precisely as:

- (A) covering a region with shapes that may overlap slightly at edges.
- (B) covering the entire plane with copies of shapes leaving no gaps and no overlaps.
- (C) covering a region while leaving small gaps for grout.
- (D) arranging shapes so their angles sum to less than 360° at each vertex.

37. To construct a 75° angle with compass and straightedge, a valid route is:

- (A) trisect a 225° angle to get 75° directly.
- (B) bisect a 100° angle.
- (C) construct 60° and 90° , bisect the 30° gap idea instead: take $60^\circ + (\text{half of } 30^\circ) = 60^\circ + 15^\circ = 75^\circ$, where 15° is half of a constructed 30° .
- (D) construct 45° and add 35° .

38. At a vertex, three regular hexagons (120° each) are placed. What is the total angle, and what does it imply?

- (A) 300° , leaving a 60° gap.
- (B) 480° , overlapping by 120° .
- (C) 240° , leaving a 120° gap.
- (D) 360° exactly, so three hexagons tile the plane perfectly around the point.

39. You are given two sides of a triangle, 8 cm and 3 cm. For the third side x , the triangle inequality restricts x to:

- (A) any value from 3 cm to 8 cm.
- (B) greater than 11 cm.
- (C) exactly 5 cm or exactly 11 cm.
- (D) greater than 5 cm and less than 11 cm (i.e. $8 - 3 < x < 8 + 3$).

40. A construction is called 'exact' (not just an accurate drawing) because:

- (A) the pencil lines are infinitely thin.
- (B) the compass measures lengths to the millimetre.
- (C) its correctness rests on geometric facts about equal radii and equidistance, so it is right in principle no matter how shaky the hand, rather than on careful measuring.
- (D) the result is traced over a printed template.

41. Equilateral triangles and squares are used in a mixed tiling. At one vertex, FOUR triangles and ONE square meet. The angle total is:

- (A) 330° , leaving a 30° gap, so this is NOT a valid vertex.
- (B) 360° , a valid vertex.
- (C) 390° , an overlap of 30° .
- (D) 300° , leaving a 60° gap.

42. Which angle is NOT constructible using only a compass and straightedge?

- (A) 30° , by bisecting a 60° angle.
- (B) 40° , because constructing it would amount to trisecting a 120° angle, which is impossible with these tools.
- (C) 15° , by bisecting a 30° angle.
- (D) 90° , by erecting a perpendicular.

43. A regular polygon with interior angle exactly 90° is the square. In a tiling using only this polygon, the number meeting at each vertex is:

- (A) three, since $3 \times 90 = 270$ looks close.
- (B) two, since two right angles make a straight line.
- (C) four, since $360 \div 90 = 4$.
- (D) six, as with triangles.

44. A triangle is to be constructed given two angles 50° and 60° and the side BETWEEN them is 6 cm. This is which construction case, and is it unique?

- (A) SAS, and it is unique.
- (B) SSA, the ambiguous case.
- (C) SSS, requiring all three sides.
- (D) ASA, and it gives a unique triangle, since the two angles and their included side fully determine it.

45. Among regular polygons, exactly three tile the plane on their own. Listing their interior angles, which set is correct?

- (A) 60° , 90° , 108° (triangle, square, pentagon).
- (B) 90° , 120° , 135° (square, hexagon, octagon).
- (C) 108° , 120° , 144° (pentagon, hexagon, decagon).
- (D) 60° (triangle), 90° (square), 120° (hexagon) — the only interior angles that divide 360° a whole number of times.

46. A right angle is constructed and bisected to give 45° . To then construct a 22.5° angle, you:

- (A) trisect the 45° angle, giving 15° pieces.
- (B) bisect the 45° angle once more, since $45 \div 2 = 22.5$.
- (C) subtract 45° from 67.5° .
- (D) bisect a 60° angle.

47. The perpendicular bisector of a segment is, in precise set terms, the set of all points that are:

- (A) the same distance from both endpoints of the segment.
- (B) exactly the midpoint of the segment.
- (C) strictly inside the segment.
- (D) closer to one chosen endpoint than the other.

48. Three different regular polygons meet at a tiling vertex: a triangle (60°), a regular hexagon, and one more polygon. If the hexagon contributes 120° , the third polygon's interior angle is 180° . What does this tell you?

- (A) The third polygon is a 'straight' polygon with a 180° angle.
- (B) The third polygon is another hexagon.
- (C) No real regular polygon has a 180° interior angle, so a triangle and a hexagon alone cannot be completed by a single third regular polygon at that vertex.
- (D) Two triangles and a hexagon would work instead, giving exactly 360° .

49. You construct an angle of 60° , then a second 60° angle sharing the same arm beside it, then a third 60° beside those. The three together form an angle of 180° . The point of this is that:

(A) three 60° angles make a right angle.

(B) 60° cannot be repeated, so the construction fails.

(C) three 60° angles fill a straight angle, which is also why six equilateral triangles (two such straight angles) close a full 360° point in the triangular tiling.

(D) three 60° angles make 360° .

50. A regular polygon is claimed to tile the plane because 'its interior angle is less than 180° '. The simplest counterexample disproving this loose rule is the:

(A) square, whose 90° angle is under 180° .

(B) equilateral triangle, whose 60° angle is under 180° .

(C) regular hexagon, whose 120° angle is under 180° .

(D) regular pentagon, whose 108° interior angle is under 180° yet $360 \div 108 = 3.33$ is not whole, so it does not tile.

51. To construct the perpendicular bisector you need two crossing points. If you construct only ONE arc crossing (above the segment) and none below, you can still draw a perpendicular by:

(A) drawing any line through the single crossing; it will be perpendicular automatically.

(B) drawing a horizontal line through the crossing.

(C) joining that single crossing to the segment's midpoint — but you must first find the midpoint, so a single crossing alone is not enough; two crossings (or a crossing plus the midpoint) are required to fix the line.

(D) doubling the arc radius.

52. A regular polygon has 9 sides (a nonagon) with interior angle 140° . At a vertex, can a whole number of nonagons tile, and if not, by how much do the nearest whole numbers miss?

(A) No; two give 280° (80° short) and three give 420° (60° over), so $360 \div 140 \approx 2.57$ is not whole and it cannot tile.

(B) Yes, three nonagons give exactly 360° . (C) Yes, two nonagons give exactly 360° .

(D) No, but two nonagons give exactly 360° with a tiny overlap.

53. A semi-regular tiling vertex uses one square (90°), one regular hexagon (120°) and two more equal regular polygons. Each of those two must have interior angle:

(A) 75° , which is impossible for a regular polygon (the smallest interior angle is 60° for a triangle, and 75° corresponds to no regular polygon), so this vertex configuration cannot be completed with two equal regular tiles.

(B) 75° , supplied by a regular polygon between a triangle and a square.

(C) 60° , two equilateral triangles. (D) 90° , two squares.

54. Two equilateral triangles (60°) and two regular hexagons (120°) are arranged around a point. Does the vertex close exactly?

- (A) No, it leaves a 60° gap.
- (B) Yes, because $2 \times 60 + 2 \times 120 = 120 + 240 = 360$ exactly.
- (C) No, it overlaps by 60° .
- (D) Yes, but only because there are four tiles.

55. A right triangle is to be constructed by RHS with hypotenuse 5 cm and one leg 5 cm. Is this possible?

- (A) Yes, it forms an isosceles right triangle.
- (B) Yes, the third side is 0 cm, which is allowed.
- (C) No, because in a right triangle the hypotenuse is strictly the longest side; a leg cannot equal the hypotenuse, so no such triangle exists.
- (D) Yes, the third side is also 5 cm, forming an equilateral right triangle.

56. Bisecting an angle locates a point equidistant from its two arms. The tool that finds such a point exactly, without any measuring, is:

- (A) a protractor set to half the angle.
- (B) a marked ruler measuring equal distances.
- (C) a set square aligned to the arms.
- (D) equal-radius compass arcs swung from the two points where an initial arc cuts the arms.

57. A polygon's interior angle is 144° (regular decagon). A student wants to combine it at a vertex with squares (90°) only to make a mixed tiling. Can one decagon plus some squares close a vertex exactly?

- (A) Yes, with two squares ($216 \div 90$ rounded to 2).
- (B) Yes, with three squares.
- (C) Yes, with 2.4 squares.
- (D) No, because $360 - 144 = 216$ and $216 \div 90 = 2.4$ is not a whole number, so no whole number of squares fills the rest.

58. The single overarching rule that decides which shapes fit around any interior vertex of a tiling is that the corner angles meeting there sum to:

- (A) exactly 180° , a straight line.
- (B) exactly 360° , the full turn — no more (which overlaps), no less (which gaps).
- (C) at most 360° , allowing small gaps.
- (D) exactly 270° , three right angles.

59. You are constructing a triangle ABC by SAS: side AB = 5 cm, side AC = 7 cm, and the included angle A = 120° . Compared with the same two sides but included angle 60° , the side BC (opposite A) will be:

- (A) the same length, since the two given sides are unchanged.
- (B) shorter when the angle is 120° , because larger angles fold the sides closer.
- (C) longer when the angle is 120° than when it is 60° , because a wider included angle spreads the far endpoints further apart.
- (D) exactly double, since 120° is double 60° .

60. A semi-regular tiling has the same arrangement of polygons at every vertex. One proposed vertex is: triangle (60°), square (90°), triangle (60°), and a regular twelve-gon

(150°). Does this single vertex close to exactly 360°?

(A) No, it leaves a 30° gap.

(B) No, it overlaps by 30°.

(C) Yes, since $60 + 90 + 60 + 150 = 360$ exactly.

(D) Yes, but only because four tiles are used.

Solutions — Mathematics (Class 7), Chapter 14: Constructions and Tilings — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	D	31	D	41	A	51	C
2	A	12	B	22	B	32	C	42	B	52	A
3	A	13	A	23	D	33	B	43	C	53	A
4	D	14	D	24	B	34	B	44	D	54	B
5	C	15	B	25	B	35	C	45	D	55	C
6	A	16	A	26	C	36	B	46	B	56	D
7	A	17	D	27	D	37	C	47	A	57	D
8	B	18	A	28	D	38	D	48	C	58	B
9	B	19	C	29	A	39	D	49	C	59	C
10	B	20	A	30	A	40	C	50	D	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) For the two equal arcs from P and Q to intersect, their radius must exceed half the segment, i.e. more than 4 cm. With a maximum of 3 cm the arcs fall short of the midpoint region and never cross, so the construction fails. The claim that any radius works ignores that arcs with radius below half the length cannot meet. The idea that merely being shorter than the full length suffices is wrong for the same reason. Requiring the radius to be exactly half would put both crossings on the segment itself, giving no line off the segment, so equality is not the rule either; it must be strictly greater than half.

2. (A) Choosing L and R equidistant from A makes A the midpoint of LR. Point T, found by equal arcs from L and R, is equidistant from L and R, hence lies on the perpendicular bisector of LR. That bisector passes through the midpoint A and is perpendicular to LR (which lies along m), so AT is perpendicular at A. Merely being a midpoint does not force perpendicularity; lines through a midpoint can have any slope. There is no 60° step in this construction. T is not the circumcentre of a triangle here; the equal-arc logic is about equidistance from L and R, not from A.

3. (A) A circle through A and B has both points at the same distance (its radius) from the centre, so the centre is equidistant from A and B. The set of all such points is precisely the perpendicular bisector of AB, so the centre lies on it. The midpoint is only one point of that bisector, not the only possible centre. Points strictly inside AB are not generally equidistant from both ends except the midpoint, and points on line AB itself are not equidistant from A and B unless they are the midpoint, which already lies on the bisector.

4. (D) The two arc crossings are each equidistant from X and Y, so the line through them is the locus of points equidistant from X and Y. By symmetry this line is perpendicular to XY, meeting it at 90° regardless of the (valid) radius used. The radius affects only how far apart the crossings are, not the angle. So 80° , an arbitrary angle, or 60° are all impossible; the angle is fixed at 90° by the geometry of equidistance.

5. (C) A triangle exists only if each pair of sides sums to more than the remaining side. Here $4 + 5 = 9$, which is less than 10, so the two shorter arcs (radius 4 and radius 5 from the ends of the 10 cm base) never reach each other and never cross. Having three different lengths does not guarantee a triangle. Comparing 10 to the sum of all three sides is meaningless; the test compares two sides to the third. There is no rule that a side cannot exceed 5 cm. So the triangle is not constructible.

6. (A) SAS needs the angle to be the included angle, i.e. the one formed where the two given sides meet. Sides 6 and 5 with the 50° angle between them fix the triangle uniquely. An angle opposite a given side is the ambiguous SSA case, not SAS, and need not determine a unique triangle. Three angles give the shape but not the size (no side length), so infinitely many triangles fit. Sides 6, 5 and 12 fail the triangle inequality ($6 + 5 = 11 < 12$) and form no triangle at all.

7. (A) The three interior angles of a triangle sum to 180° . Two given angles must therefore sum to less than 180° to leave a positive third angle. Here $110^\circ + 80^\circ = 190^\circ > 180^\circ$, so the two arms drawn from the ends of the base diverge and never meet on the same side; no triangle forms. A negative third angle is geometrically impossible. A positive base length does not rescue an impossible angle pair. ASA does not require the two angles to be equal; it requires their sum to be under 180° .

8. (B) RHS gives a right angle, the hypotenuse and one side; the right triangle is determined. The missing leg satisfies $\text{leg}^2 + 5^2 = 13^2$, so $\text{leg}^2 = 169 - 25 = 144$, giving $\text{leg} = 12$ cm. Adding the hypotenuse and leg (18) or subtracting them (8) ignores the right-angle relationship. The hypotenuse is the longest side and is not required to equal a leg, so the triangle is perfectly constructible.

9. (B) SSS, SAS and ASA each fix a single triangle up to congruence. The case where you know two sides and an angle that is opposite one of them (SSA) can sometimes be satisfied by two different triangles, because the swinging side can meet the base line in two places.

So SSA is the non-unique case, while the other three are valid congruence conditions that pin down exactly one triangle.

10. (B) The exterior angle equals $180^\circ - \text{interior} = 180^\circ - 156^\circ = 24^\circ$. Since the exterior angles of a regular n -gon sum to 360° , each is $360^\circ/n$, so $n = 360 \div 24 = 15$. Dividing 156 by 13 has no geometric meaning here. Guessing from 'between a decagon and dodecagon' ignores the exact exterior-angle method. The exterior angle is $180 - \text{interior}$, not interior minus 132, so 24 sides is wrong.

11. (C) Dividing the arc into three equal chord-pieces does NOT divide the angle into three equal angles, because equal chords on a circle subtend equal angles only when measured as arcs from the centre, and the naive 'three arcs' trick does not achieve that for a general angle. It is a known impossibility result that a general angle cannot be trisected with these two tools, even though it can always be bisected. The claim that three equal arcs split the angle is exactly the trap. There is no special rule favouring angles over 90° . Bisection certainly does work without a protractor, so the last option is false too.

12. (B) An angle is determined by the chord PQ cut from an arc of fixed radius centred at the vertex. By using the same radius and transferring the chord PQ, the new arc subtends an identical chord, so the new angle equals the original. No protractor is involved; the compass carries length, not degree readings. The arc colour is irrelevant. Stepping off PQ once reproduces the angle directly; there is no doubling-then-halving.

13. (A) While $90 + 90 + 0$ does total 180° , a triangle needs three strictly positive angles. A 0° angle collapses two of the sides onto each other, so the figure degenerates into a line and is not a triangle. The fact that the sum hits 180° is necessary but not sufficient. There is no genuine 'thin right triangle' with a true 0° vertex. The sum 180° is in fact exactly the required total, so the last option misstates the rule.

14. (D) Since F is equidistant from G and H (both lie on the arc of radius $FG = FH$), F lies on the perpendicular bisector of GH. That bisector meets m at the midpoint of GH, and the segment from F to this midpoint is perpendicular to m , giving the foot of the perpendicular. G is merely one arc crossing, not the foot. Not every point of m is the foot; only one point gives a perpendicular. The foot relates to F, not to the point nearest G.

15. (B) A regular polygon self-tiles only when its interior angle divides 360° a whole number of times, so that an exact whole number of corners fills the full turn with no gap or overlap. Here $360 \div 162 \approx 2.22$ is not a whole number, so the polygon cannot tile by itself. Rounding to 2 leaves a wedge ($2 \times 162 = 324 < 360$), so two do not fit. You cannot have a fractional number of tiles at a vertex. Being 'close to 180° ' is irrelevant; closeness does not make a tiling.

16. (A) Tiles meeting at a vertex must fill exactly 360° , with each contributing an angle A , so a whole number k of them gives $kA = 360^\circ$, meaning $360 \div A$ is a positive integer. 'Less

than 120° is wrong: the hexagon tiles with $A = 120^\circ$, not under it. 'Multiple of 45° ' would exclude the triangle (60°) and hexagon (120°), which do tile. Requiring only that $360 \div A \geq 2$ admits non-integer results like 2.5, which leave gaps or overlaps.

17. (D) One dodecagon contributes 150° , leaving $360 - 150 = 210^\circ$ for a SINGLE additional polygon. But every regular polygon has interior angle strictly less than 180° , so no single polygon can supply 210° ; the gap must be filled by two or more polygons instead. A second dodecagon (150°) plus the first totals 300° , not 360° . A triangle (60°) gives 210° , a square (90°) gives 240° , neither equalling 360° with just the dodecagon — and the question asks for ONE polygon, which is impossible.

18. (A) Adding the corner angles: $90 + 120 + 150 = 360^\circ$, the full turn, so the vertex closes exactly with no gap and no overlap. There is no leftover wedge (a gap would mean the sum is under 360) and no piling-up (an overlap would mean over 360). Being multiples of 30° is incidental; what matters is the exact sum of 360° , which here it achieves.

19. (C) Two octagons contribute $2 \times 135 = 270^\circ$, leaving $360 - 270 = 90^\circ$ to fill exactly. A 90° interior angle belongs to a square, giving the well-known octagon-and-square tiling. A triangle (60°) or hexagon (120°) would not equal the required 90° , leaving a gap or overlap. There is no regular polygon with a 45° interior angle, and 'cutting a corner in half' is not a regular tile.

20. (A) Three pentagons cover $3 \times 108 = 324^\circ$, leaving $360 - 324 = 36^\circ$. A regular polygon's interior angle is always at least 60° (the equilateral triangle, the smallest), so 36° is impossible; the gap cannot be plugged by any single regular polygon. An equilateral triangle has a 60° interior angle, not 36° . There is no 'pentagon half' tile. The leftover is 36° , not 0° , so three pentagons do not tile.

21. (D) The perpendicular bisector relies on EQUAL radii from both ends, which makes both crossings equidistant from X and Y. With unequal radii (5 vs 6) the crossings are no longer equidistant from X and Y, so the line tilts and misses the midpoint; it is neither perpendicular nor a bisector. Both crossings existing is not enough — they must come from equal radii. A line through two such crossings is not parallel to XY, nor does it become an angle bisector of X.

22. (B) Angles sum to 180° , so $A = 180 - (40 + 30) = 180 - 70 = 110^\circ$. Because $40 + 30 = 70$ is under 180° , the two arms from B and C converge to a point and the triangle exists; ASA does not require the base angles to be equal. The third angle is 110° , not 70° (that is just the sum of the two given angles). There is no rule that the base must be the longest side.

23. (D) For a regular n-gon with $n > 6$, the interior angle is more than 120° (the hexagon's value) but always less than 180° . So $360 \div (\text{interior angle})$ lies strictly between $360/180 = 2$ and $360/120 = 3$, i.e. between 2 and 3, and can never be the whole number needed to

close a vertex. Interior angles never reach 180° for a real polygon, so that option is wrong. Physical size is irrelevant to tiling. The exterior angle stays positive (it is $360/n$), so it never goes negative.

24. (B) Start at 60° . First bisection gives 30° , second gives 15° , third gives 7.5° . That is three halvings of the original 60° , so the smallest piece is 7.5° . Stopping at 15° counts only two halvings. Dividing 60 by 6 is not what bisection does. 3.75° would require a fourth halving, one more than described.

25. (B) Tiling demands the angles meeting at a vertex sum to EXACTLY 360° . Two corners of 165° give 330° , a 30° shortfall — a real, visible gap, not harmless. Rounding 2.18 up to 3 gives $3 \times 165 = 495^\circ$, an overlap of 135° , also not a tiling. Merely being under 180° does not let a polygon tile; only an exact whole-number division of 360° does, which 165° fails.

26. (C) One hexagon uses 120° , leaving $360 - 120 = 240^\circ$. Each triangle supplies 60° , so $240 \div 60 = 4$ triangles close the vertex. Dividing 240 by 120 wrongly uses the hexagon angle for triangles. Three triangles give only 180° , not the needed 240° . Six triangles (360°) would leave no room for the hexagon, overshooting.

27. (D) 105° splits as $60^\circ + 45^\circ$. A 60° angle is directly constructible, and 45° is got by bisecting a constructed right angle (90°), so 105° is constructible. The set $60 + 30 + 20$ sums to 110, not 105, and 20° is not even constructible. $90 + 30 = 120$, not 105. Trisecting a general angle is impossible with these tools, so the last route fails.

28. (D) Sum the corner angles: two squares give 180° , three triangles give 180° , total 360° , so the vertex closes with no gap or overlap. There is no shortfall (which would be a gap) and no excess (which would be an overlap). The number of tiles being five is incidental; what matters is that the angle sum is exactly 360° .

29. (A) Take any triangle and rotate a copy 180° about the midpoint of one side; the two together make a parallelogram, and parallelograms tile by translation. So every triangle tiles the plane, not just the equilateral one. The condition about angles dividing 360° applies to a single REGULAR polygon meeting itself corner-to-corner, not to triangles in general, whose three different angles (summing to 180° , so two sets give 360°) combine cleverly around a vertex. Right triangles tile too, but so do all the others, so 'only right triangles' is wrong.

30. (A) A valid tiling vertex needs exactly 360° . A total of 350° is 10° short, so an uncovered wedge (gap) remains and the tiling is invalid. 'Close enough' is the classic trap — only exact 360° works. A shortfall is a gap, not an overlap (an overlap comes from totals over 360°). Totals under 360° are perfectly possible; they simply indicate gaps, so no remeasuring is forced.

31. (D) Because $BP = BQ$ (same arc radius) and $PR = QR$ (equal arcs from P and Q), triangles BPR and BQR are congruent, so angle PBR = angle QBR; thus BR bisects angle ABC. Merely being the midpoint of PQ does not by itself guarantee equal angles unless the arms are equal as set up here, but the bisection reason is the equal-angle congruence, not just a midpoint. R is not fixed at 30° from each arm — the half-angle depends on the original angle. R is not a circumcentre; the construction is about equal radii.

32. (C) If three corners fill 360° , each interior angle is $360 \div 3 = 120^\circ$, which is the regular hexagon. A 60° angle would need six corners ($k = 6$), not three. A 90° angle needs four corners ($k = 4$). A 108° angle does not divide 360° evenly at all, so it cannot tile with any whole k .

33. (B) The known corners total $150 + 60 + 60 = 270^\circ$, leaving $360 - 270 = 90^\circ$ for the fourth tile, which is a square. A third triangle (60°) would total only 330° . A hexagon (120°) would overshoot to 390° . A second 12-gon (150°) would overshoot to 420° . Only 90° closes the vertex exactly.

34. (B) A point on the perpendicular bisector of a side is equidistant from that side's two endpoints. The crossing of two such bisectors is therefore equidistant from all three vertices, so a circle centred there passes through all three — the circumcentre. Equidistance from the three SIDES (not vertices) defines the incentre, which comes from ANGLE bisectors, not perpendicular bisectors. The crossing is generally not on any side at all, and angle bisectors give a different point (the incentre).

35. (C) Interior = $180 - \text{exterior} = 180 - 40 = 140^\circ$. For self-tiling, 360 must be a whole-number multiple of 140° , but $360 \div 140 \approx 2.57$, not an integer, so it cannot tile. Rounding down to 2 leaves $2 \times 140 = 280^\circ$, a gap of 80° . The interior angle is 140° , never 320° (interior angles are under 180°). Three corners give 420° , an overlap, so 3 does not work either.

36. (B) A tiling covers the whole plane exactly: every point is covered, with no gaps (uncovered spots) and no overlaps (double-covered spots). Allowing overlaps violates the definition, as does leaving gaps. Requiring vertex angles to be under 360° actually describes a gap, the opposite of a tiling — at every interior vertex the angles must equal exactly 360° .

37. (C) $75^\circ = 60^\circ + 15^\circ$. A 60° angle is constructible directly; 15° is half of 30° , which is half of 60° , so 15° is constructible by two bisections. Adding them gives 75° . Trisecting a general angle is impossible with these tools, ruling out the 225° route. A 100° angle is not constructible with compass and straightedge, so bisecting it is moot. 35° is not constructible, so $45^\circ + 35^\circ$ fails.

38. (D) Three hexagons give $3 \times 120 = 360^\circ$, the full turn, so they tile exactly with no gap or overlap. A total of 300° would be two hexagons plus a triangle, not three hexagons.

480° would be four hexagons (an overlap). 240° would be only two hexagons. Three hexagons land precisely on 360° .

39. (D) The third side must be more than the difference and less than the sum of the other two: $8 - 3 = 5 < x < 8 + 3 = 11$. So $5 < x < 11$. Restricting x between 3 and 8 has no basis. Values over 11 break the sum condition (the two shorter sides could not reach). At exactly 5 or 11 the triangle degenerates into a straight line (sides just touch), so those endpoints are excluded, not allowed.

40. (C) A geometric construction is justified by reasoning — equal radii produce equidistant points, which force midpoints, perpendiculars and bisectors exactly. The truth of the result does not depend on measuring with a ruler or protractor, only on the logic. Lines having thickness is a real-world imperfection, not the reason for exactness. The compass carries equal lengths but does not 'measure to the millimetre' the way the option claims. Tracing a template is not a construction at all.

41. (A) Four triangles give $4 \times 60 = 240^\circ$, plus one square 90° , totals 330° . That is 30° short of 360° , so a gap remains and the vertex is invalid. It does not reach 360° , so it is not valid. It does not exceed 360° , so there is no overlap. 300° would be five triangles or three triangles plus a square, not this mix.

42. (B) A 40° angle would require trisecting a constructible 120° angle, and general trisection is impossible with compass and straightedge, so 40° is not constructible. By contrast, 30° comes from bisecting the constructible 60° , 15° from bisecting that 30° , and 90° from a standard perpendicular construction — all genuinely constructible.

43. (C) Squares fill 360° with $360 \div 90 = 4$ corners at each vertex. Three squares give only 270° (a 90° gap). Two squares give 180° (a straight line, not a closed vertex). Six is the count for the 60° triangle, not the 90° square.

44. (D) Two angles with the side lying between them is the ASA condition, which determines a unique triangle. It is not SAS (that needs two SIDES and the included angle). It is not the ambiguous SSA case (two sides and a non-included angle). It is not SSS, which needs three sides; here only one side is given.

45. (D) Self-tiling regular polygons are the equilateral triangle (60° , six per vertex), square (90° , four per vertex) and regular hexagon (120° , three per vertex); $360 \div$ each is a whole number. The pentagon (108°) gives 3.33, not whole, so it does not tile. The octagon (135°) gives 2.67, also not whole. The decagon (144°) gives 2.5, again not whole. So only the 60/90/120 set works.

46. (B) Bisecting 45° halves it to 22.5° , which is constructible. Trisecting 45° is impossible with these tools and would in any case give 15° , not 22.5° . Subtracting from 67.5° requires already having 67.5° , which is not a basic constructible angle here. Bisecting 60° gives 30° , not 22.5° .

47. (A) The perpendicular bisector is the locus of points equidistant from the two endpoints — that is its defining property and why it is perpendicular through the midpoint. The midpoint is just one point of this infinite line. Points strictly inside the segment lie on the segment, not on the (perpendicular) bisector except at the midpoint. Points closer to one endpoint lie off the bisector, on one side of it.

48. (C) One triangle (60°) plus one hexagon (120°) total 180° , leaving $360 - 180 = 180^\circ$ for a single third polygon. But a regular polygon's interior angle is always strictly less than 180° , so no such polygon exists; you cannot finish the vertex with one more tile. There is no 'straight 180° polygon'. A second hexagon would give $60 + 120 + 120 = 300^\circ$, not 360° . (Note two triangles and a hexagon give $60 + 60 + 120 = 240^\circ$, also not 360° , so that option is false as stated.)

49. (C) Three 60° angles sum to 180° , a straight angle. Doubling that (six 60° angles) gives 360° , which is exactly why six equilateral triangles meet at each vertex in the triangular tiling. Three 60° angles make 180° , not a 90° right angle. A constructed 60° can certainly be copied repeatedly. Three of them give 180° , not 360° ; it takes six to reach 360° .

50. (D) The pentagon has interior angle $108^\circ < 180^\circ$ yet $360 \div 108 \approx 3.33$ is not an integer, so it leaves a gap and does NOT tile — a clean counterexample to the loose 'under 180° ' claim. The square (90°), triangle (60°) and hexagon (120°) all DO tile, so none of them disproves the claim; only the pentagon does.

51. (C) A single point does not determine a line; you need two points. The perpendicular bisector is fixed by two equidistant crossings, or equivalently by one crossing together with the known midpoint. A line through one crossing in an arbitrary direction is generally not perpendicular. A 'horizontal' line has no meaning without orientation and is not guaranteed perpendicular. Doubling the radius just relocates the crossing; it still leaves you with one point.

52. (A) Interior angle 140° means $360 \div 140 \approx 2.57$, not a whole number. Two corners give $2 \times 140 = 280^\circ$ (80° short, a gap), three give $3 \times 140 = 420^\circ$ (60° too much, an overlap). Neither closes the vertex, so the nonagon cannot self-tile. Claims that two or three give exactly 360° contradict these sums.

53. (A) After the square (90°) and hexagon (120°), the remaining angle is $360 - 90 - 120 = 150^\circ$, split equally between two tiles, so each needs 75° . No regular polygon has a 75° interior angle (the sequence is $60^\circ, 90^\circ, 108^\circ, 120^\circ, \dots$), so the configuration fails. Two triangles (60° each) give 120° , making a total of $90 + 120 + 120 = 330^\circ$, not 360° . Two squares (90° each) give $90 + 120 + 180 = 390^\circ$, an overlap. So neither alternative closes the vertex; the required 75° simply does not exist.

54. (B) Two triangles give 120° , two hexagons give 240° , total 360° , so the vertex closes with no gap or overlap (this is the well-known trihexagonal-type vertex). There is no shortfall and no excess. The number of tiles being four is incidental; the exact 360° sum is what matters.

55. (C) By Pythagoras, $\text{hypotenuse}^2 = \text{leg}^2 + (\text{other leg})^2$, so the hypotenuse exceeds each leg. If a leg equalled the hypotenuse ($5 = 5$), the other leg would be $\sqrt{(25 - 25)} = 0$, collapsing the triangle into a segment — not a genuine triangle. An isosceles right triangle has two equal LEGS, not a leg equal to the hypotenuse. A 0 cm side is degenerate, not allowed. There is no 'equilateral right triangle'.

56. (D) Equal-radius arcs from the two arm-crossings meet at a point equidistant from both arms, giving the bisector by pure equidistance — no measurement. A protractor measures degrees (and you would need to know half the angle first), which is measuring, not constructing. A marked ruler measures lengths, again measuring. A set square only gives fixed angles like 90° , 45° , 30° , not a bisector of an arbitrary angle.

57. (D) One decagon uses 144° , leaving $360 - 144 = 216^\circ$. Filling this with squares needs $216 \div 90 = 2.4$ squares, not a whole number, so no exact closure with squares. Two squares give 180° , totalling 324° (a 36° gap). Three squares give 270° , totalling 414° (an overlap). A fractional 2.4 squares is not physically possible. So the vertex cannot be closed this way.

58. (B) Around a point there are 360° of turning, so the tile corners meeting at an interior vertex must total exactly 360° — under it leaves a gap, over it forces an overlap. 180° is only a straight angle, not a full turn. 'At most 360° ' permits gaps, which violate the tiling definition. 270° is just three right angles and does not fill the plane around a point.

59. (C) With two fixed sides hinged at A, opening the included angle wider pushes the far ends B and C further apart, so BC grows as angle A increases; thus 120° gives a longer BC than 60° . Keeping AB and AC fixed does NOT keep BC fixed — BC depends on the angle between them. A larger angle spreads, not folds, the endpoints, so BC is not shorter. BC does not scale linearly with the angle, so it is not exactly double; the relationship is geometric, not proportional to the degree measure.

60. (C) Add the corner angles: $60 + 90 + 60 + 150 = 360^\circ$, the full turn, so the vertex closes with no gap or overlap. There is no shortfall (a gap) and no excess (an overlap). The count of four tiles is incidental; what matters is the exact 360° sum, which it reaches.